

§2.2: Sandwiched Singularities, Extended Graphs, and Decorated Germs

Nishtha Tikalal

August 23, 2026

1. PROPOSITION IV (AS SEEN FROM THE PAPER)

Proposition 1.1. *Let $(X, 0)$ be a rational singularity with reduced fundamental cycle, and $\mathcal{C}$ a plane curve germ corresponding to $(X, 0)$. If $\mathcal{C}$ has smooth branches, the number of branches is given by $\text{mult } X - 1$.*

2. DEFINITIONS

Definition 2.1. Let $(X, 0) \subset (\mathbb{C}^N, 0)$ be a surface singularity. The *multiplicity* $\text{mult } X$ is defined geometrically as the number of intersections $\#X \cap L$ of X with a generic complex $(N - 2)$ -dimensional affine subspace $L \subset \mathbb{C}^N$ passing close to the origin. For rational singularities, multiplicity is a topological invariant computed from the resolution graph by the formula:

$$\text{mult } X = -Z_{\min}^2$$

The components of this definition are broken down as follows:

- $(X, 0) \subset (\mathbb{C}^N, 0)$: A complex surface inside an N -dimensional complex space with an isolated singularity at the origin.
- $\text{mult } X$: A geometric measure of the folding or density of the surface at the singular point (a smooth point has a multiplicity of 1).
- L : A generic complex linear subspace of codimension 2. Its intersection with the 2-dimensional surface X yields a finite, countable set of isolated points.
- **“Passing close to the origin”**: Perturbing L slightly away from the origin unfolds the singularity. This allows the distinct generic intersection points ($\#X \cap L$) to be counted cleanly.
- $Z_{\min}$: The fundamental (minimal) cycle. This is the unique minimal positive integer combination of exceptional curves $Z = \sum a_i E_i$ satisfying $Z \cdot E_i \leq 0$ for all components E_i in the resolution graph.
- $-Z_{\min}^2$: The negative self-intersection of the fundamental cycle. Because the resolution intersection matrix is negative definite, $Z_{\min}^2$ is negative. Flipping the sign yields a positive integer matching the geometric intersection count.

For a branch of a plane curve germ $\mathcal{C}$ to be *smooth*, it must satisfy the following conditions:

- **No discontinuities**: The branch is topologically connected and forms a continuous path through the origin.
- **No sharp edges or kinks**: Geometrically, the branch possesses a well-defined, continuously turning tangent vector at every point, including at the origin.
- **Analytic smoothness**: Algebraically, each individual branch can be parameterized locally by a single smooth complex parameter $t \mapsto (x(t), y(t))$ such that the derivative vector is non-vanishing, meaning the branch is locally isomorphic to a non-singular complex disk.

3. MATHEMATICAL OUTPUT

Artin's Theorem and the Genus Formula

To prove our result about the minimal resolution graph G , we need to check the arithmetic genus.

Theorem 3.1 (Artin's Characterization Theorem). *Let $(X, 0)$ be a normal surface singularity, and let Z be its fundamental cycle on a good resolution $\tilde{X}$. The singularity is rational if and only if the arithmetic genus of the fundamental cycle is zero:*

$$p_a(Z) = 0$$

Proposition 3.2 (The Arithmetic Genus Formula). *For any effective cycle $Y = \sum_{i=1}^n c_i E_i$ on a resolution surface, its arithmetic genus $p_a(Y)$ is calculated from the graph data by:*

$$p_a(Y) = 1 + \frac{1}{2}Y^2 + \frac{1}{2} \sum_{i=1}^n c_i (2g(E_i) - 2 - E_i^2)$$

Proof of Proposition 2.4

Proof. Let $(X, 0)$ be a rational surface singularity. We are given that its fundamental cycle $Z_{\min}$ is reduced. Recall that being reduced means all the coefficients c_i in the cycle are exactly 1:

$$Z_{\min} = \sum_{v \in G} E_v$$

where G is the minimal resolution graph with n vertices. Recall that since our exceptional curves E_v are rational curves, their geometric genus is zero ($g(E_v) = 0$) for every vertex $v \in G$.

Now we plug $c_v = 1$ and $g(E_v) = 0$ into the arithmetic genus formula for $Z_{\min}$:

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 + \frac{1}{2} \sum_{v \in G} (2(0) - 2 - E_v^2)$$

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 + \frac{1}{2} \sum_{v \in G} (-2 - v \cdot v)$$

$$p_a(Z_{\min}) = 1 + \frac{1}{2}Z_{\min}^2 - n - \frac{1}{2} \sum_{v \in G} (v \cdot v)$$

Because $(X, 0)$ is a rational singularity, Artin's theorem states that $p_a(Z_{\min}) = 0$. Let's set the equation to zero and multiply everything by 2 to clear the fractions:

$$0 = 2 + Z_{\min}^2 - 2n - \sum_{v \in G} (v \cdot v) \implies Z_{\min}^2 = \sum_{v \in G} (v \cdot v) + 2n - 2$$

Recall that for any tree graph with n vertices, the sum of all vertex valences $d(v)$ is exactly equal to twice the number of edges. Since a tree has $n - 1$ edges, the sum of the valences is $2(n - 1) = 2n - 2$. We can substitute $\sum_{v \in G} d(v) = 2n - 2$ into our formula:

$$Z_{\min}^2 = \sum_{v \in G} (v \cdot v) + \sum_{v \in G} d(v) = \sum_{v \in G} (v \cdot v + d(v))$$

Next, we look at the conditions from **Proposition 2.2**. The plumbing graph G' for the smooth point is made by adding m branches as (-1) leaf vertices to G . Proposition 2.2 tells us that:

1. There is exactly one special vertex $v'_0 \in G'$ where $v'_0 \cdot v'_0 + a(v'_0) = -1$.
2. For all other vertices $v' \neq v'_0 \in G'$, $v' \cdot v' + a(v') = 0$.

When we go from G' back to our minimal graph G by removing the m leaf branches, the vertex valences change:

- For any vertex $v \neq v'_0$, no branches were attached there, so the valence in G is the same as in G' ($d(v) = a(v)$). This means $v \cdot v + d(v) = 0$.
- For the special vertex v'_0 , all m branches were attached here, so its valence drops by m in the minimal graph ($d(v'_0) = a(v'_0) - m$).

Now we can compute the sum for $Z_{\min}^2$ using these two facts:

$$Z_{\min}^2 = \sum_{v \neq v'_0} (0) + (v'_0 \cdot v'_0 + d(v'_0))$$

$$Z_{\min}^2 = v'_0 \cdot v'_0 + a(v'_0) - m$$

Using the condition $v'_0 \cdot v'_0 + a(v'_0) = -1$ from Proposition 2.2, this simplifies directly to:

$$Z_{\min}^2 = -1 - m$$

Recall that the multiplicity of a rational singularity is defined as $\text{mult } X = -Z_{\min}^2$. Therefore:

$$\text{mult } X = -(-1 - m) = m + 1$$

If we rearrange this equation to solve for the number of branches m , we get:

$$m = \text{mult } X - 1$$

This completes the proof. □

REFERENCES

- [1] R. E. Gompf and A. I. Stipsicz, *4-Manifolds and Kirby Calculus*, Graduate Studies in Mathematics, vol. 20, American Mathematical Society, Providence, RI, 1999.
- [2] O. Plamenevskaya and L. Starkston, *Unexpected Stein fillings, rational surface singularities, and plane curve arrangements*, Geometry & Topology, 27(3), 2023, pp. 1083–1166. Preprint available at arXiv:2006.06631 [math.GT], <https://arxiv.org/abs/2006.06631>.
- [3] G.-M. Greuel, C. Lossen, and E. Shustin, *Introduction to Singularities and Deformations*, Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2007.
- [4] T. de Jong and D. van Straten, *Deformation theory of sandwiched singularities*, Duke Mathematical Journal, 95(3), 1998, pp. 451–522.
- [5] W. P. Barth, K. Hulek, C. A. M. Peters, and A. Van de Ven, *Compact Complex Surfaces*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, Vol. 4, Springer-Verlag, 2nd enlarged edition (2004).
- [6] A. Beauville, *Complex Algebraic Surfaces*, London Mathematical Society Student Texts 34, Cambridge University Press, 2nd edition (1996).